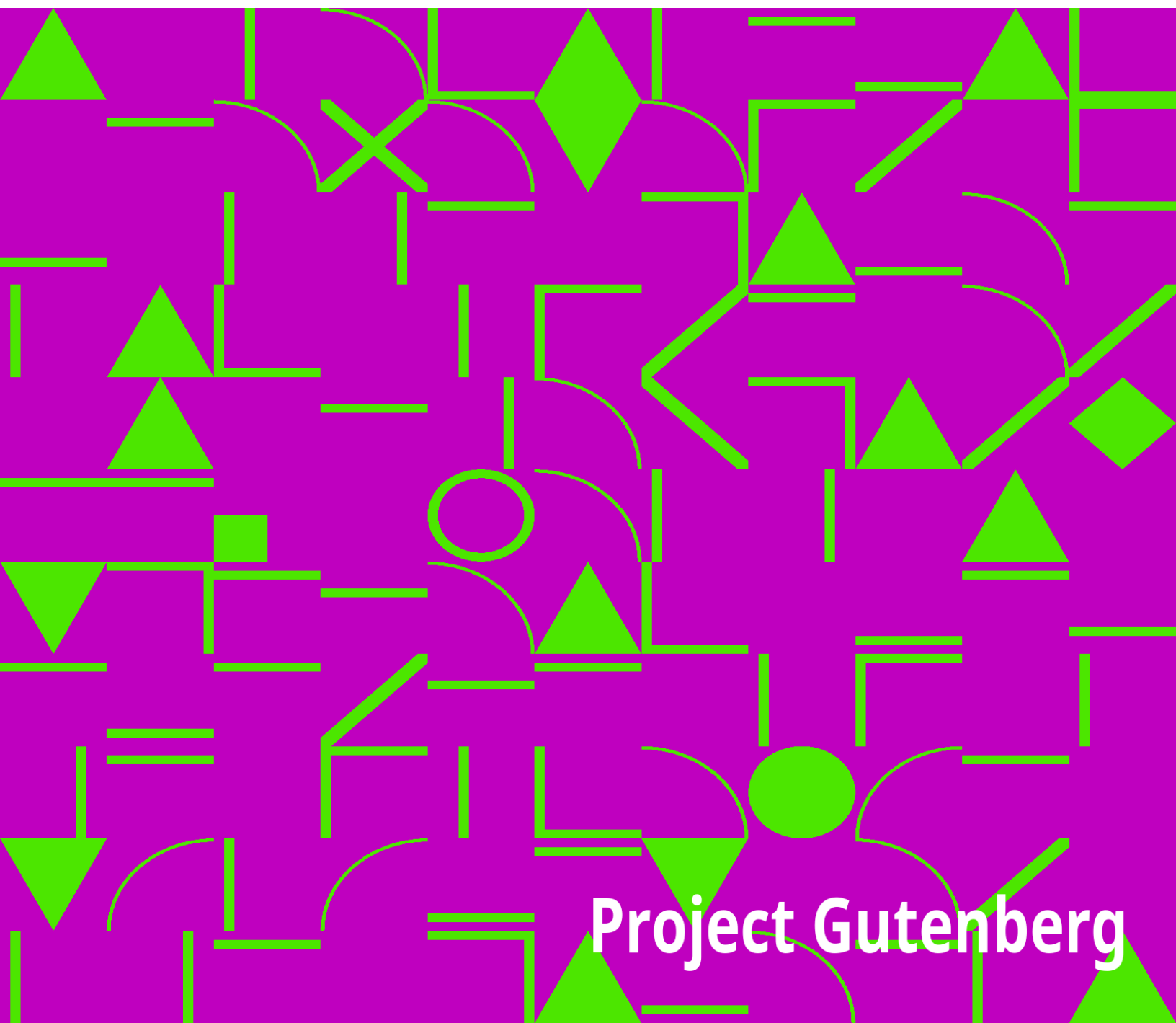


A Bilateral Division of the Parietal Bone in a Chimpanzee; with a Special Reference to the Oblique Sutures in the Parietal

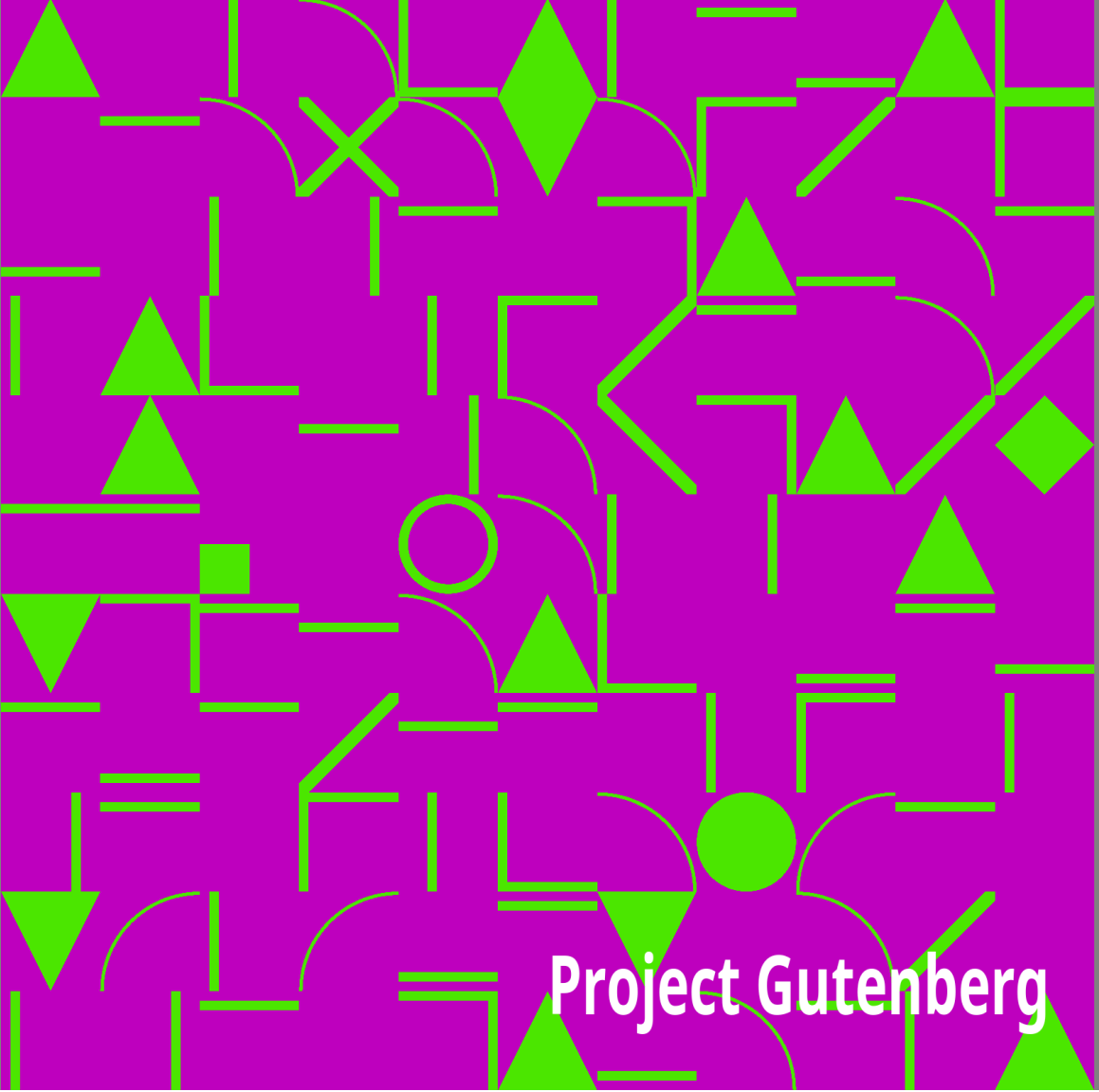
Aleš Hrdlička



Project Gutenberg

A Bilateral Division of the Parietal Bone in a Chimpanzee; with a Special Reference to the Oblique Sutures in the Parietal

Aleš Hrdlička



Project Gutenberg

The Project Gutenberg eBook of A Bilateral Division of the Parietal Bone in a Chimpanzee; with a Special Reference to the Oblique Sutures in the Parietal

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: A Bilateral Division of the Parietal Bone in a Chimpanzee; with a Special Reference to the Oblique Sutures in the Parietal

Author: Aleš Hrdlička

Release date: October 19, 2010 [eBook #34101]
Most recently updated: January 7, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/34101

Credits: Produced by Larry B. Harrison and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK A BILATERAL
DIVISION OF THE PARIETAL BONE IN A CHIMPANZEE; WITH A
SPECIAL REFERENCE TO THE OBLIQUE SUTURES IN THE
PARIETAL ***

***A Bilateral Division of the Parietal
Bone
in a Chimpanzee; with a Special
Reference
to the Oblique Sutures in the
Parietal.***

By ALEŠ HRDLIČKA

AUTHOR'S EDITION, extracted from BULLETIN
OF THE

American Museum of Natural History,

VOL. XIII, ARTICLE XXI, pp. 281–295.

New York, Dec. 31, 1900.

The Knickerbocker Press, New York

Article XXI.—A BILATERAL DIVISION OF THE PARIETAL BONE IN A CHIMPANZEE; WITH SPECIAL REFERENCE TO THE OBLIQUE SUTURES IN THE PARIETAL.

By ALEŠ HRDLIČKA.

The first to describe a case of division of the parietal bone in apes was Johannes Ranke, in 1899. [\[1\]](#) The skull in question is that of an adolescent female orang, one of 245 orang crania in the Selenka collection in the Munich Anthropological Institute. The abnormal suture divides the right parietal into an upper larger and a lower smaller portion. "The suture runs nearly parallel with the sagittal suture," but, as the illustration shows ([Fig. 1](#)), it descends in its posterior extremity towards the temporo-parietal suture, and terminates in this a few millimetres in front of the lambdoid suture. The abnormal suture shows but little serration, and the articulation of the two divisions of the parietal bone is squamous in character, the lower portion overlapping the upper. Below the junction of the abnormal with the coronal suture, the latter takes a pronounced bend forward. A similar bend in the coronal suture is present in the same specimen on the left side. This is common among the other orang skulls in the collection. The portions of the coronal suture below and above the bend differ somewhat in character.

Besides the above-mentioned complete division, Ranke found among the 245 orang skulls 13 with incomplete division of the parietal bone. The division consisted invariably of a longer or shorter remnant of a horizontal "parietal suture," ending in the coronal suture at the top of the bend above referred to. A similar anterior remnant of an abnormal parietal suture was found by Ranke in a young chimpanzee skull; but the author questions the

word "chimpanzee," which evidently means that the identity of the skull is somewhat doubtful.

In consequence of his finds, Ranke believes both complete and incomplete divisions in the parietal bone to be much more frequent in the orang than in man. [2] He also thinks that the bend usually present in the coronal suture in the orang signifies that, "even where there are no traces of a parietal suture, such a suture has actually existed in an earlier stage of development." This implies the development of the adult parietal bone in the orang from two original segments, one above the other.

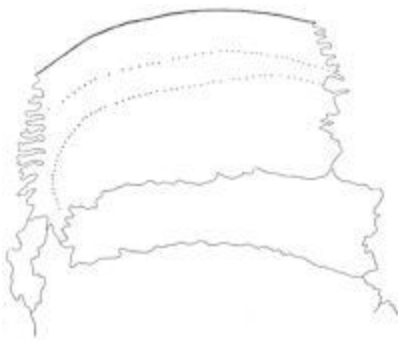


Fig. 1. Division of the Right Parietal in an Orang (Ranke, Abh. d. k. bayer. Akad. d. Wiss., II cl., XX Bd., ii Abth.).

The divisions which I am about to describe occur, one in each parietal, in the skull of a nine-year-old male chimpanzee, which was captured, when young, in West Africa. Later on he was one of the attractions of the Barnum and Bailey Circus, and was familiarly known as Chico. The chimpanzee died in 1894, since when his skin and bones have been preserved in the American Museum of Natural History, New York City. Prof. J. A. Allen, the curator of the Zoölogical Department of the Museum, has kindly given

me permission to describe the skeletal parts for publication. [3]

The most interesting part of Chico is unquestionably the skull. The divisions of the parietal bones which the specimen presents are not only the first complete divisions of the parietal observed in a chimpanzee, but are also unique in character, no divisions of the same nature having been observed before, either in man, in apes, or in monkeys. The position and extent of the divisions in this skull will throw considerable light on the question of the aberrant, complete divisions of the parietal bone, by which term may be designated divisions differing from the typical horizontal ones.

The skull under consideration shows in general a good development and an almost perfect symmetry. The capacity of the brain cavity, measured according to Flower's method, is 390 c.c.

The masculine features of this skull, and particularly the temporal ridges, are not quite as marked as those of another skull of an adolescent male

chimpanzee in the Museum. The temporal ridges are slightly prominent, and in their middle third, over part of the frontal and the parietal bones, not more pronounced than in some human crania. They are, however, situated very high. Their upper lines or boundaries touch each other over a part of the sagittal suture, a little back of the bregma; while the lower lines approach to within 6 mm. of the sagittal suture. The supraorbital ridges are not very massive, although prominent to such a degree that, when the skull rests on the occipital condyles and on the teeth, the plane of the orbits is almost vertical. The sagittal crest is insignificant; the occipital crest is high, but not very massive. The zygomatic arches are less strong than they are in an average white male; and the mastoids are small, even smaller than in an average adult white female.

The second dentition is incomplete; the third molars have not reached the level of the opening of their sockets. The condition of the sutures, so far as their patency is concerned, does not bear the same relation to the stage of dentition as it does in man: all the sutures of this skull are more or less obliterated. There are no signs on any part of the skull that point to the closure of any of the sutures as premature. In detail, the condition of the sutures is as follows: The speno-maxillary articulation is completely closed, but still plainly traceable. Of the various facial sutures, only remnants are open; the suture in the zygomatic arch, however, is almost fully patent on both sides. The speno-frontal articulation is completely obliterated on the left, but traces of it remain on the right side. The left temporo-sphenoidal and squamo-frontal sutures (the squama of the temporal articulates with the frontal bone) are, with the exception of the basal part of the former, which remains open, quite obliterated, but on the right side both are open. The temporo-parietal sutures, with the exception of 8 mm. of the anterior end of the suture on the right side, are both entirely closed and hardly traceable. The coronal suture is partly open on the left, and wholly open on the right, up to a point a little below the middle of the anterior border of the parietal bone. At this point on each side, the lower portion of the coronal suture bends backward and continues as the anomalous suture; the upper portion of the coronal, particularly on the right, is completely obliterated, though still traceable. There are no signs left of the sagittal and lambdoid sutures, and only the basal portions of the

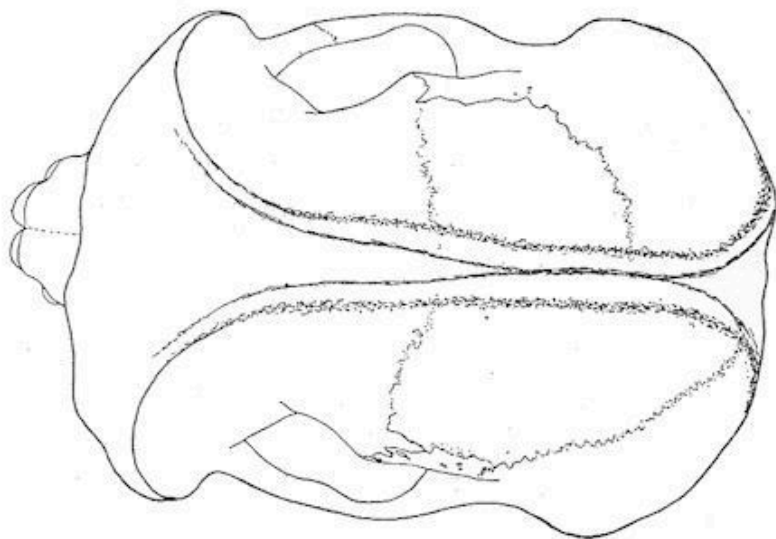
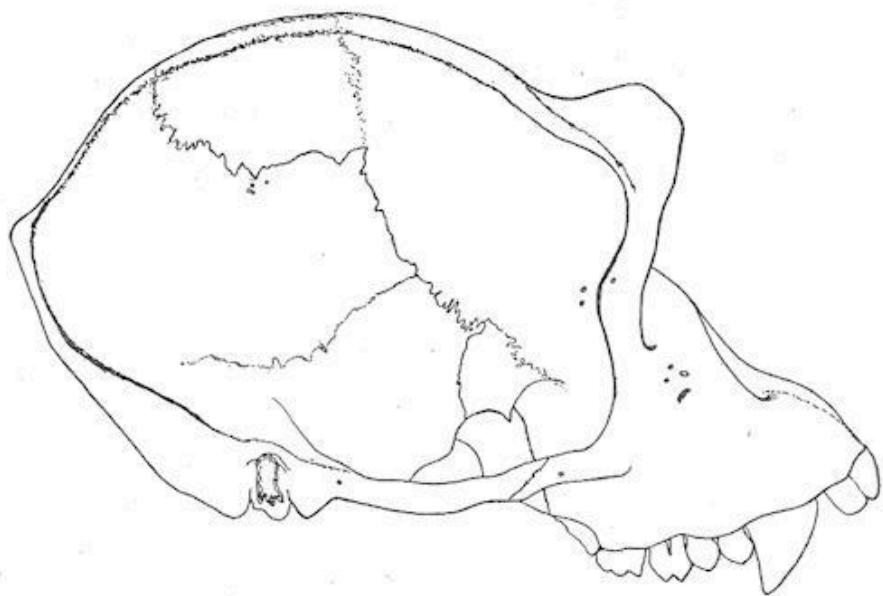
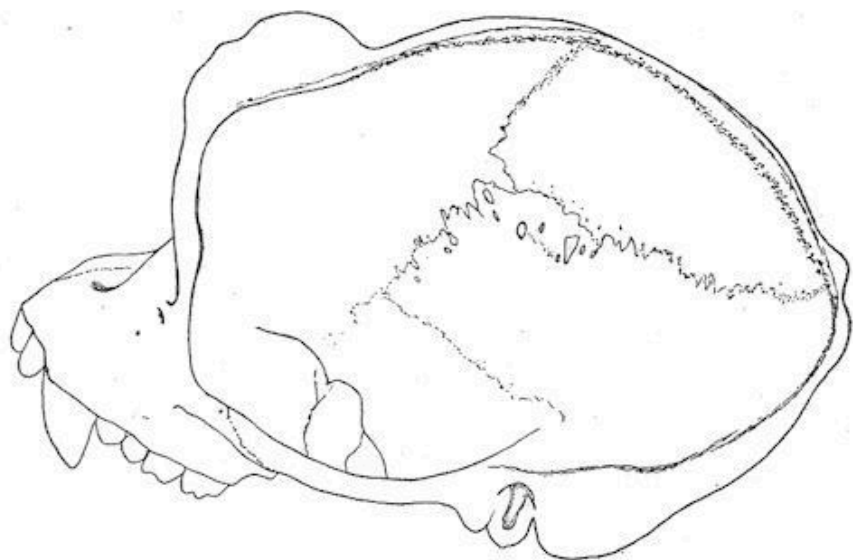
temporo-occipital articulation remain. The palatine sutures, also, are entirely obliterated.

The skull shows no important anomalies besides the division of the parietals.

The divisions of the parietal bones begin on the left 32 mm., on the right 28 mm. (measured with a tape), above the point of junction of the coronal and temporo-parietal sutures. From the point where the anomalous sutures leave the coronal suture, to the bregma, the distance on the left is 44 mm., on the right 42 mm. The excess of size of the left over the right parietal bone along the coronal suture (6 mm.) compensates the greater height of that portion of the right temporal squama which articulates with the frontal bone. Measured across their middle from the temporo-parietal suture, the two parietals appear to be almost of equal size (left 82 mm., right 80 mm.). In an antero-posterior direction, from the beginning of the division to the middle of the parietal portion of the occipital crest, both bones measure the same, namely 75 mm.

The division in the left parietal begins at a V-shaped cleft, which is filled with a process of the frontal bone. There are slightly distinct markings on the bone and a number of insular ossicles, which make it probable that the cleft had been originally much greater and was largely filled by a Wormian or, rather, a fontanel bone, the lower border of which has subsequently united with the parietal.

For 30 mm. from its beginning the abnormal suture proceeds directly backward, and to this extent shows but little obliteration. The original cleft has, it seems, extended up to this point. From here the suture takes a slight bend upwards, and proceeds almost directly upwards and backwards, becoming gradually obliterated, until it disappears at the temporal ridge, 16 mm. from the median line. Originally the suture must have terminated on the posterior border of the parietal bone, not far from the lambda. The whole suture shows fairly good serration. The coronal suture on this side, below the division, shows serration about equal to that of the abnormal suture; the obliterated portion above this was, so far as can be seen, more simple.



Figs. 2–4. Skull of an Adolescent Male Chimpanzee.

On the right side the division of the parietal may also have begun with a cleft in the anterior border of the bone, but, owing to the advanced state of obliteration of the upper portion of the coronal suture on this side, the existence of the cleft cannot be fully ascertained. Here also the abnormal suture, at first wholly open, runs for the first 26 mm. directly backwards; at this point the suture, still quite patent, takes a turn somewhat sharper than that on the left, and proceeds for 16 mm. backwards and upwards; here it takes a second turn, and proceeds almost directly upwards towards the sagittal suture. This last portion of the abnormal suture is considerably obliterated, and on and beyond the temporal ridge is scarcely traceable. The point at which the division has reached the sagittal suture is situated a little behind the middle of the latter. The abnormal as well as the open part of the coronal suture on this side shows a simpler serration than the corresponding sutures on the left side.

In this specimen there is on neither side any encroachment of the lower portion of the parietal bone upon the frontal, such as Ranke lays stress on in the case of his orangs. A second skull of an adolescent male chimpanzee, in the Museum of Natural History, has a decided bend in the coronal suture, not unlike that which Ranke describes, and which, as he thinks, generally indicates an old parietal division; but in this case the bend is situated between the inferior and superior boundaries of the prominent temporal ridge, and apparently owes its origin to the latter ([Figs. 2, 3, 4](#)).

The main interest in the case just described centres in the direction of the abnormal sutures, and in the clearness with which the two divisions appear as equivalent and of the same origin, although one divides the parietal completely, while the other is restricted to one of its angles.

As to the course of the abnormal suture in the parietal bone, in all the cases thus far reported, the division runs in a horizontal direction (cases of Tarin, Soemmering, Gruber, Hyrtl, Welcker, Turner, Putnam, Dorsey, Ranke, and others); or it runs obliquely from or near the middle of the lambdoid suture to some part of the temporo-parietal suture, the sphenoidal angle, or the lower portion of the coronal suture (cases of Curnow, Ekmark, Gruber,

Hyrthl, Lucae, Welcker, Putnam, Traquair, Ranke); in a case of *Simia silenus* described by Gruber and in an Egyptian cranium described by Smith, the divisions run to the lambda and begin respectively slightly above the pterion and at it. In Boyd's and in two of Hyrthl's cases, the abnormal suture begins at or below the bregma on the coronal margin of the parietal bone, and ends at or near its mastoid angle; finally, in Blumenbach's (cited by Welcker), Bianchi's, Fusari's, and Coraini's cases (those of Coraini include two monkeys) the division is vertical, passing between the temporo-parietal and sagittal sutures. The left division in our chimpanzee approaches those in Gruber's *Simia silenus* and Smith's cases; but it originates much higher anteriorly, and terminates slightly below the lambda on the occipital border of the parietal. The division in the right parietal of the chimpanzee, beginning slightly below the middle of the anterior border of the bone, and ending slightly back of the middle of its sagittal border, has no analogy among the cases previously described.

The difference in extent and terminations of the two abnormal sutures in the chimpanzee is of particular interest in connection with the problem of the significance and origin of those divisions of the parietal bone that involve more or less only one of its angles.

Since the observations of Toldt, ^[4] and more recently of Ranke, ^[5] on the development of the parietal bone in the human embryo, it appears, though it cannot as yet be said whether the fact is or is not general, that the bone originates from two centres of ossification. These centres appear in most cases one directly above the other, but, as Ranke himself shows, ^[6] and as can hardly be otherwise, these primitive components of the parietal do not always show the same relations in size or position. The centres blend together, ordinarily, at the end of the third or during the first half of the fourth month of foetal life. On this account, the typical, complete, horizontal division of the human parietal bone, when met with at any time after the fourth month of foetal life, is generally interpreted to-day as a retardation of the union, or a persistence of separation, of the two original segments of the bone. Opinion, however, is still unsettled as to the significance of the more atypical, oblique divisions of the parietal, particularly of those where the separation is limited to one angle. Up to the recent contribution on the subject by Ranke, the weight of opinion on the point, although rather briefly expressed, seems to have been in favor of attributing to these smaller,

oblique divisions, the same significance as was given to the more typical, horizontal ones. Gruber, [7] in reporting a new case of a bilateral oblique suture in the parietal bone, calls the separated mastoid angles "the secondary posterior parietals." Hyrtl and Welcker advance no definite theories on this point, though the latter expresses an opinion [8] that in both the horizontal division and the separation of the mastoid angle of the parietal bone the development of the condition may be identical. In 1883 Prof. F. W. Putnam, in describing one of his Tennessee skulls with an abnormal oblique suture in each parietal, [9] referred the development of the separated mastoid angle on the right side, as well as the larger oblique inferior portion of the parietal on the left side, to a "separate centre" of ossification. Ranke [10] opposes both Gruber's and Putnam's opinion, and presents instead a theory somewhat vague and not satisfactorily demonstrated, by which he accounts for the origin of oblique sutures from partial horizontal sutures in the parietal bone through "half-pathological processes." In his words, "the oblique parietal suture is allied to the half-pathological conditions of the skull; it is wholly unjustifiable to speak, as W. Gruber has done, of a separate *Parietale secundarium posterius*, severed by the suture, as of a typical, in a certain sense normal, formation. The oblique parietal suture is nothing more than an incomplete (posterior), true, *i. e.*, typical, parietal suture with a sagittal course, modified by certain half-pathological conditions." These half-pathological conditions are produced, the author explains on the preceding page, "durch Einknickung der nach Herrn G. H. Meyer 'plastisch' aufwärts gebogenen hinteren Scheitelbeinränder."

This opinion of Ranke calls for a few words about the incomplete horizontal parietal sutures. These sutures are apparently very rare in human adults, only five instances being on record (4 Ranke's, 1 Turner's). They are more frequent in orangs (Ranke), and quite common (as Ranke shows, and as I found independently before Ranke's publication of his observations) in the human embryos near term and in new-born or very young infants. In the human family, these partial divisions of the parietal generally begin in the posterior part, and run sagittally to the posterior border of the bone, ending in this border at or near its middle. In orangs the incomplete horizontal divisions seem to begin, as a rule, in the anterior part, and end at or near the middle of the anterior border of the parietal. The length of these divisions

varies from a few millimetres to several centimetres, and they even reach up to the centre of the parietal bone. ^[11] These divisions are, without doubt, the remains of the original anterior and posterior clefts, or, if we go a step further, of the original intervening antero-posterior space between the original inferior and superior segments of the parietal. From the very first contact of the growing centres, the median extremity of these clefts is bounded both below and above by a mass of bone; and when the anterior or posterior border of the parietal comes finally in contact with the frontal or occipital bone, the anterior and posterior sagittal clefts, if they still exist, lie between two well-developed, firm portions of the bone. Under these circumstances it is quite impossible to imagine any disturbance, mechanical or pathological, that could affect solely or mainly the median portion of the cleft, and cause a deflection downward in this portion of the division, or cause its extension to the inferior border or even the anterior-inferior angle of the parietal.

There are only two factors that can possibly affect and modify the course of the incomplete parietal suture, and both of these would show their influence mainly or entirely on the distal portion of the same. These two factors are, first, an abnormal development, either defective or excessive, of one of the original parietal segments; and, secondly, influences that would interfere with the freedom of full growth of the anterior or posterior border of the parietal.

In the first case, as can easily be imagined or even artificially demonstrated, there would be possible only a lower or higher situation or an obliquity affecting mostly the marginal portion of the division. The results would be low or high sagittal sutures, and curved or oblique sutures diverging from the parietal eminence,—effects entirely different from the actually observed oblique sutures that sever the lower portion of the parietal, or its mastoid angle.

Influences interfering with the free development of the anterior or posterior border of the parietal bone could only deflect upwards or downwards the marginal end of an incomplete parietal suture, or, at most, in a case of a short suture, render it oblique or curved in its entirety. No pathological condition, unless it were accompanied by a fracture, could extend even a

deflected antero-posterior incomplete division to any of the borders of the bone.

There are, it seems to me, only three possible ways in which an oblique suture, extending between any two borders of the parietal bone, can be produced.

In the first case the oblique suture, or rather a suture-like formation, may be the effect of an early fracture. A fracture produced in adult life is generally recognizable as such; but a fracture dating from earlier stages of life, produced before the growth of the bone has ceased, may, if not entirely obliterated, present more or less the characteristics of a suture. I have seen several skulls where a division in the parietal bone or the temporal squama presented at the same time features of a fracture and suture; in one or two of these cases so much so, that it was and still is impossible for me to decide exactly which of the two conditions I had before me. Gruber describes one such case [\[12\]](#) as an instance of an oblique parietal suture, while Hyrtl and Ranke both consider this case as one with an acquired division. To differentiate a congenital real oblique suture from a division which is the result of a fracture, we must be guided largely by the situation, form, and serration of the division, and the condition of the surrounding bones, especially that of the opposite parietal. A straight course, ending with one extremity in or near the middle of the anterior or posterior border of the parietal, a complex serration, no continuity of the division on the neighboring bones, and particularly a co-existence of an allied or similar division on the opposite parietal,—all favor the conclusion that the division under consideration is a real congenital suture, and not the result of a fracture.

In the second case there are reasons for believing that an oblique suture of the parietal bone can originate in the same way as the horizontal one, namely, through a persistence of the original separation between the two centres from which the bone is developed, and a co-existent difference in the relative position or the relative growth of the two centres. It is in this connection that the above-described division in the parietals of the chimpanzee will prove of value.

The occasional persistence of the separation between the two original segments of the parietal bone is sufficiently demonstrated by the presence of the complete horizontal parietal suture. Differences in the relative position of these segments can be observed in a limited degree in Ranke's illustrations of embryos, before referred to; it can be deduced from such cases as the two of Hyrtl, [\[13\]](#) in which the division of the parietal was directed from the upper portion of the anterior to the lower portion of the posterior border of the bone. The most pronounced change in the position of these centres may be witnessed in cases where the parietal bone shows a perfect vertical instead of a horizontal suture. Such cases have been referred to before, and I presented at the meeting of the Association of American Anatomists, in 1899, several such examples, found by me in skulls of monkeys in Professor Huntington's anatomical collection in the Medical Department of Columbia University. One of these specimens is shown in the accompanying illustration ([Fig. 5](#)).

A difference in the relative growth of the two centres of the parietal bone is well shown in the difference of size between the inferior and superior portions of the parietal in cases of the complete horizontal suture in the same. In the majority of such cases on record the superior portion is larger, particularly anteriorly, than the inferior; so much so, that that condition seems to be the typical one. The difference in the size of the two portions of the parietal, and in their relative anterior and posterior height, is most pronounced in one of Gruber's cases, [\[14\]](#) where the "parietal suture" begins only 10 mm. above the pterion, and ends 40 mm. above the asterion. In Dorsey's case [\[15\]](#) the lower portion of the divided parietal is 12 mm. higher than the upper. The same condition as is found in Gruber's case, here mentioned, exists in the almost identical left division of the second case of Putnam, of which I have a photograph in my hands. A somewhat similar excess of the posterior over the anterior part of the lower severed portion of the parietal can also be seen in the illustrations of the cases of Tarin, Lucae, and Turner (Admiralty Islands skull). In Calori's interesting case [\[16\]](#) there is a decided excess of the lower portion of the divided parietal in its posterior portion on the left and in its anterior portion on the right side.

In case the upper segment was not vertically above the lower one, but in a position a little more forward or backward of it; and, furthermore, if the

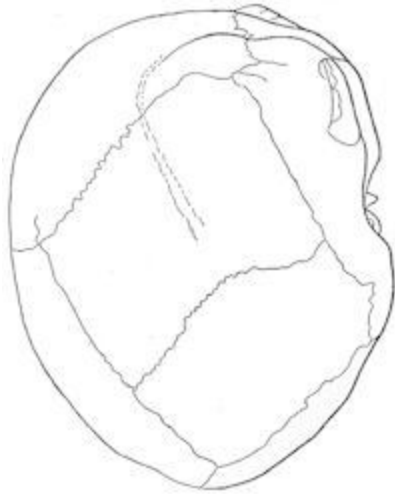


Fig. 5. *Macacus rhesus* (Medical Department, Columbia University), showing a Complete Division of the Right Parietal Bone in a Vertical Direction.

relative growth of the two segments differed, and their separation remained permanent,—the separation of any portion of the parietal bone in almost any form and to almost any extent might result. Such coincidence of anomalous conditions, although necessarily rare, cannot, from what we know on the subject in parietal and other bones, be declared improbable. All cases where oblique suture on one side co-exists with more or less horizontal suture on the other side in the parietal bone, as in the second of Putnam's cases, would of course point directly to a similar origin of the anomaly on both sides of the cranium. That such cases have not been more frequently observed is largely due, I think, to the rarity of bilateral parietal divisions.

A third mode of development of the oblique suture in the parietal bone suggests itself where the severed portion of the bone is small, and that is the possible existence of a supernumerary, third centre of ossification. I am by no means ready to defend this theory, yet there are cases in which it would afford the easiest explanation. I have a Peruvian skull at hand, in which there is a bilateral, quite symmetrical quadrangular separate piece of bone, encroaching on the mastoid process of the parietal. The surface of the left parietal bone in this skull measures across its middle in antero-posterior direction 120 mm., in infero-superior direction 130 mm.; similar measures of the right parietal are respectively 117 and 130 mm. The separate bone on the left measures across its middle in antero-posterior direction 20 mm., in infero-superior direction 12 to 21 mm.; the same portion on the right measures respectively 25 and 11 to 15 mm. Both pieces are joined to the parietal bone by a squamous suture ([Fig. 6](#)).

It is apparent that the separate pieces of bone in this case are too small to be easily taken for representatives of one of the regular centres of ossification of the parietal bone; but the same pieces are somewhat too large, and especially too singularly outlined and joined to the parietal, to be without difficulty diagnosed as simple Wormian or fontanel bones. One of Ranke's cases, ^[17] though the separation of the mastoid angle is oblong instead of

quadrangular, as in the Peruvian skull, seems to me to present a similar difficulty in properly diagnosing the nature of the severed portion. This group of cases needs further observation, particularly on the bones of infants and embryos. I have two monkey skulls at hand which actually show a multiplicity of the original segments of the parietal. These specimens will be described in a future publication.

So much as to the *formation* of the oblique sutures in the parietal. It should not be forgotten that such sutures can be *simulated* by those which divide true Wormian or fontanel bones from the parietal. The distinction between the real oblique parietal and these extra-parietal sutures must depend largely on the extent of the division and form of the separate piece of bone.

We may now return to the skull of our chimpanzee. In considering the nature of the divisions in the parietal bones of this skull, we can at once and absolutely discard the idea of the divisions being due to fractures, or being boundaries of Wormian or fontanel bones, and thus really extra-parietal in their nature. There is nothing about the sutures, or the divided pieces, or the neighboring bones, that would even suggest such an explanation; and in our records on Wormian and fontanel bones we find no analogies either in man, or apes, or lower animals, to the conditions here observed. The necessary conclusion from this can only be that we have before us two examples of real parietal division.

The division on the left side, had it existed alone, would be readily acceptable as an instance of the "parietal suture." The anterior extremity and more than the anterior third of the course of the division correspond exactly to the same features of a typical, horizontal "parietal suture;" while the elevation of the posterior extremity of the division, though unusual, can readily be explained as due to an excess in growth of the inferior original centre of the bone, which may, in addition, have been situated slightly posterior to the upper centre.

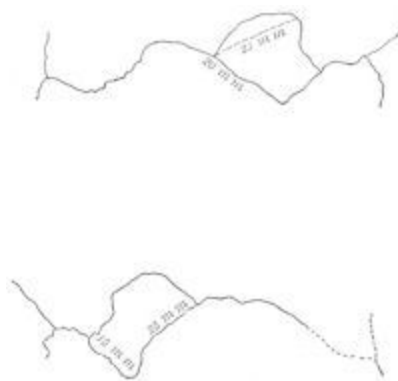


Fig. 6 (99/3550). Quadrilateral Fontanel Bones in a Peruvian Male Skull, encroaching upon the Mastoid Angle of the Parietals.

The division in the right parietal of the chimpanzee begins at its anterior end, and runs for the first third of its course in the same way as that on the left side; its posterior end, however, does not reach the lambdoid, but turns up and ends in the sagittal border. Should this formation have existed alone, I should be inclined to consider it either as the result of an accessory centre of the parietal, or, possibly, as a persistence of the anterior portion of the divided superior centre of the bone, the posterior portion of the same being united with the lower segment of the parietal in the usual way. With the division of the left parietal in the same skull before me, everything points to a similar origin of the division on both sides, and to the right as well as the left division being a true "parietal suture," deflected less on the left and more on the right side by a disproportion in growth of the two original, regular segments of each of the bones.

The disproportion of growth of the two original segments of the parietal bone will, I believe, be found more common as attention is directed to this subject. It can be well explained, though there may at times be other factors present, by a difference in the blood-supply to the two centres. This of course may occur not only in different skulls, but also on the two sides of the same cranium.

Footnotes

[1] Die überzähligen Hautknochen des menschlichen Schädeldachs, Abh. d. k. bayer. Akad. d. Wiss., II Cl., XX Bd., II Abth., pp. 36 et seq., Fig. 17.

[2] *L. c.*, p. 41. Among 3000 Bavarian crania, Ranke found but one with complete and three with incomplete parietal sutures; basing his conclusion on this observation, he says, "Bei den Orangutanschädeln ist die Häufigkeit der Scheitelbeinnäthe circa 40 mal grösser als bei dem erwachsenen Menschen."

[3] Since finding the abnormal sutures on this skull, I have been able to present the same at a meeting of the Association of American Anatomists (1899) and before the Ethnological Society of New York City (1900).

[4] Toldt, C., in Maska's Hdb. d. gerichtl. Med., 1882, v. III, p. 515; the same in his U. d. Entwick. d. Scheitelbeins d. Menschen, Zeitschr. f. Heilkunde, 1883, v. IV, pp. 83–86.

[5] *L. c.*, pp. 324–330.

[6] *L. c.*, pp. 327–330, Figs. 29–32.

[7] Gruber, W., *Beobacht. a. d. menschl. u. vergl. Anat.*, Berlin, 1879, II Heft, pp. 12–15.

[8] Welcker, H., *Untersuch. ü. d. Wachsthum u. Bau d. menschl. Schädels*, Leipzig, 1862, p. 109.

[9] Putnam, F. W., *Abnormal Human Skulls from Stone Graves in Tennessee* (*Proc. A. A. A. S.*, XXXII, 1884, p. 391).

[10] *L. c.*, p. 309.

[11] Ranke's Fig. 25, p. 318.

[12] Virchow's *Archiv*, 1870, v. 50, p. 113.

[13] Hyrtl, J., *Die doppelten Schläfenlinien d. Menschenschädel*, etc. (*Denkschr. d. math. naturw. Classe d. k. Akad. d. Wiss. zu Wien*, 1871, v. XXXII, pp. 39–50).

[14] Gruber, W., *Beobacht. a. d. menschl. u. vergl. Anat.*, Berlin, 1879, II Heft, p. 15.

[15] Dorsey, G. A., *Chicago Med. Recorder*, v. XII, Feb., 1897.

[16] Calori, Luigi, *Sut. soprannum. d. Cranio Umano* (*Mem. d. Accad. d. Sc. d. Ist. d. Bologna*, 1867, pp. 327 et seq., Fig. 4).

[\[17\]](#) *L. c.*, p. 303, Fig. 13.

PUBLICATIONS

OF THE

American Museum of Natural History

The publications of the American Museum of Natural History consist of the 'Bulletin,' in octavo, of which one volume, consisting of about 400 pages, and about 25 plates, with numerous text figures, is published annually; and the 'Memoirs,' in quarto, published in parts at irregular intervals.

The matter in the 'Bulletin' consists of about twenty-four articles per volume, which relate about equally to Geology, Palæontology, Mammalogy, Ornithology, Entomology, and (in the recent volumes) Anthropology.

Each Part of the 'Memoirs,' forms a separate and complete monograph, with numerous plates.

MEMOIRS.

Vol. I (not yet completed).

PART I.—Republication of Descriptions of Lower Carboniferous Crinoidea from the Hall Collection now in the American Museum of Natural History, with Illustrations of the Original Type Specimens not heretofore Figured. By R. P. Whitfield. Pp. 1–37, pll. i-iii. September 15, 1893. Price, \$2.00.

PART II.—Republication of Descriptions of Fossils from the Hall Collection in the American Museum of Natural History, from the report of Progress for 1861 of the Geological Survey of Wisconsin, by James Hall, with Illustrations from the Original Type Specimens not heretofore Figured. By R. P. Whitfield. Pp. 39–74, pll. iv-xii. August 10, 1895. Price, \$2.00.

PART III.—The Extinct Rhinoceroses. By Henry Fairfield Osborn. Part I. Pp. 75–164, pll. xiiia-xx. April 22, 1898. Price, \$4.20.

PART IV.—A complete Mosasaur Skeleton. By Henry Fairfield Osborn. Pp. 165–188, pll. xxi-xxiii, with 15 text figures. October 25, 1899.

PART V.—A Skeleton of Diplodocus. By Henry Fairfield Osborn. Pp. 189–214, pll. xxiv-xxviii, with 15 text figures. October 25, 1899. Price of Parts IV and V, issued under one cover, \$2.00.

Vol. II. Anthropology (not yet completed).

The Jesup North Pacific Expedition.

PART I.—Facial Paintings of the Indians of Northern British Columbia. By Franz Boas. Pp. 1–24, pll. i-vi. June 16, 1898. Price, \$2.00.

PART II.—The Mythology of the Bella Coola Indians. By Franz Boas. Pp. 25–127, pll. vii-xii. November, 1898. Price, \$2.00.

PART III.—The Archæology of Lytton, British Columbia. By Harlan I. Smith. Pp. 129–161, pl. xiii, with 117 text figures. May, 1899. Price, \$2.00.

PART IV.—The Thompson Indians of British Columbia. By James Teit. Edited by Franz Boas. Pp. 163–392, pll. xiv-xx, with 198 text figures. April, 1900. Price, \$5.00.

PART V.—Basketry Designs of the Salish Indians. By Livingstone Farrand. Pp. 393–399, pll. xxi-xxiii, with 15 text figures. April, 1900. Price, 75 cts.

Vol. III. Anthropology (not yet completed).

PART I.—Symbolism of the Huichol Indians. By Carl Lumholtz. Pp. 1–228, pll. i-iv, with 291 text figures. May, 1900. Price,

\$5.00.

BULLETIN.

Volume	I, 1881– 86	Price, \$5.50
"	II, 1887– 90	" 4.75
"	III, 1890– 91	" 4.00
"	IV, 1892	" 4.00
"	V, 1893	" 4.00
"	VI, 1894	" 4.00
"	VII, 1895	" 4.00
"	VIII, 1896	" 4.00
"	IX, 1897	" 4.75
"	X, 1898	" 4.75
"	XI, Part I, 1898	" 1.25
"	II, 1899	" 2.00
"	XII, 1899	" 4.00

For sale by G. P. PUTNAM'S SONS, New York and London;
J. B. BAILLIÈRE ET FILS, Paris; R. FRIEDLÄNDER & SOHN, Berlin;
and at the Museum.

Transcriber's Note

- Footnotes have been moved to the end of the article.

*** END OF THE PROJECT GUTENBERG EBOOK A BILATERAL
DIVISION OF THE PARIETAL BONE IN A CHIMPANZEE; WITH A
SPECIAL REFERENCE TO THE OBLIQUE SUTURES IN THE
PARIETAL ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.